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MODULE 2: Biological databases

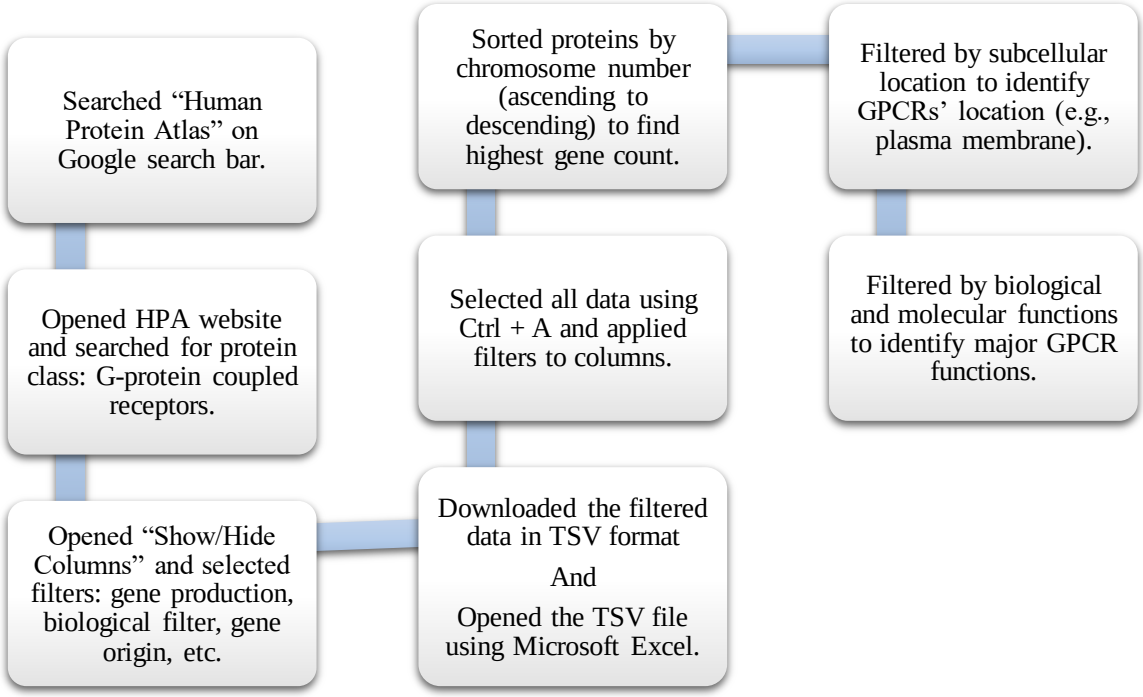
Assignment title: Protein Classification and Analysis Using the Human Protein Atlas

PROTEIN CLASS: GPCRs

Introduction

GPCRs are one of the largest families of membrane proteins in the human genome. They are involved in transmitting a wide range of signals into cells and helping in internal responses and forwarding internal responses connecting cells. These receptors respond to stimuli like hormones, neurotransmitters, and also sometimes secondary messengers. These proteins are also nowadays the target of 30 to 40% of medicinal drugs for personalized medicine therapy, targeted drug involvement, or targeted drug therapy.

Methodology



Observations

A. Chromosome Origin:

Based on the observations sorted in column F, reveals a clear distribution pattern. The findings are as follows:

1. Chromosome 3 contains the highest number of GPCR genes, with a total of three.
2. Chromosomes 5, 12, and X each carry two GPCR genes.
3. Chromosomes 4, 6, 7, 10, and 19 each have a single GPCR gene.

B. Subcellular Location:

From analysis of the subcellular location data in column L , the following observations were made:

1. The plasma membrane was the most frequently appearing location with three proteins, which matches the expected location for GPCRs.
2. Neoplasm and cytosol were also found as locations for three proteins each.
3. Rarer subcellular locations were also observed included actin filaments and nuclear speckles.
4. Although we typically think of GPCRs as being bound to the membrane for transmembrane signalling, this data shows they can also be found in nuclear and cytoplasmic regions. This opens up the possibility of mechanisms like dual localization or internal trapping within somatic compartments. Ultimately, it suggests these proteins may have more complex, compartment-specific signalling jobs than we traditionally thought.

C. Molecular and Biological Function

1. <u>Biological Process</u> : The data for specific biological processes in column I was very limited. Only one protein, associated with the gene CALY, had a listed function: endocytosis and also is a Target for FDA approved drugs.

2. <u>Molecular Function</u> : The molecular functions listed in column J confirmed the primary role of these proteins. They consistently showed receptor and signal transducer activity, which means they act as the cell's signal detectors, translating external messages into internal action.

D. Protein Class

A review of the protein classes in column H reveals a few key groups:

1. The largest group, with 12 entries, consisted of ‘predicted membrane proteins’ this is the expected class for most GPCRs.
2. A smaller group of 4 proteins was classified as ‘Predicated intracellular proteins.
3. Other categories were also present. For instance, GCR156 was labelled as a transporter along with some others and also others were marked as plasma proteins.

E. Patterns and Trends

1. The GPCR genes are most concentrated on chromosome 3, although they are also distributed across many other chromosomes.
2. Most proteins are membrane-bound as expected, but a surprising number also show dual locations in the cytosol and nucleus, suggesting they may have multiple cellular roles.
3. Their primary function is consistently shown to be receptor and signal transducer activity, while biological process data was limited to endocytosis.

Conclusion

The analysis of these 14 GPCR proteins showed a few key things. Most of their genes are on Chromosome 3, and the proteins themselves are primarily found on the cell membrane where they act as receptors and transducers.

The interesting part was finding some of these proteins inside the cell, which is unusual and suggests they might have extra jobs we don't know about yet. It was also clear that the functional data is incomplete, as most biological processes weren't listed. Overall, the findings support the known signalling role of GPCRs but also hints some interesting exceptions that are worth exploring further.